Mohammed Aayan Pathan

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Work Experience

Junior Engineering Associate — Full-Stack Web Developer Oct 2025 — Apr 2026
VOXA (voxa.club) *Dubai, UAE (Remote)*

- Refactored the Voxa MVP codebase into production-ready version 1.0, modernizing architecture with redefined user stories and responsive design.
- Co-developed investor-facing Demo app deployed at Lisbon WebSummit 2025 using React (Vite) and Node.js; responsible for version control and deployment workflow.
- Assisted in onboarding 10+ developer interns: authored a 15+ page developer wiki, provided code reviews, and enforced Git workflow standards.
- Lead migration by transitioning v1.0 authentication and data layer from Firebase to Supabase (PostgreSQL) to reduce technical debt and improve schema scalability.
- Awarded Star Associate of the Year, the highest individual recognition at VOXA.

Product and App Development Intern July 2025
VOXA (voxa.club) *Dubai, UAE (Remote)*

- Built a multi-step onboarding flow with Firebase Auth (email, Google, and GitHub sign-in), proactively initialized a structured user object from OAuth data prior to account creation to enable personalized state from first login.
- Developed a card-based practice system integrating AssemblyAI transcription with OpenAI GPT feedback generation; each session injected the transcription alongside card metadata into a structured prompt returning categorized feedback including filler word detection.

Engineering Projects

Full-Stack Developer, FormulaMetric (formulametric.com) May 2025 — Present

- Designed full-stack F1 analytics platform using React, Node.js, Express, and MySQL with normalized schema, foreign keys, and indexed lookups.
- Implemented Jolpica API data ingestion with a recursive exponential backoff algorithm to handle rate limits.
- Built regression, correlation, and sigmoid-based performance metrics for multi-season driver comparisons.
- Sorted drivers by metric factors using random-pivot Quicksort, then selected top-ranked drivers via an O(n) min-heap.
- Built interactive UI with MUI X gauges, charts, and responsive layouts supporting comparison of up to 3 drivers.

Developer, UniMetric (syntaxsnipes.github.io/UniMetric) June 2026

- Built a client-side university ranking tool processing 1,500+ institutions across 5 years of QS World University Rankings data, parsed and normalized from raw Excel exports into JSON.
- Implemented a proportional weight normalization algorithm preserving QS's relative metric ratios across any active factor subset, ensuring consistent re-ranking regardless of selected combination.
- Developed per-metric trend charts across 5 years and head-to-head comparison mode for up to 3 universities simultaneously, exposing trajectories absent from QS's own platform.
- Implemented shareable URL encoding of selected filters, year, and country for persistent custom rankings.

Team Leader, Full-Stack Developer, TLSearch, NASA Space Apps Challenge (tlsearch.org) Oct 2025 — Mar 2026

- Built a semantic search engine for NASA space-biology datasets using vector embeddings and cosine similarity ranking.
- Developed Next.js frontend with TailwindCSS and SEO-ready metadata connected to the FastAPI backend.
- Implemented FastAPI backend computing in-memory similarity scores and returning ranked JSON results.

Education

American University of Sharjah Aug 2026 — May 2030
BSc Computer Science *Sharjah, UAE*

- Awarded highest Merit Scholarship (20%), granted to applicants with 95%+ average (A*A*AA in A-Levels Achieved)
- ICSC Bronze Finalist w/honors (top 8% internationally) — International Computer Science Competition

Skills

- **Languages:** TypeScript, JavaScript, Python, SQL (MySQL, PostgreSQL), HTML, CSS
- **Frameworks & Libraries:** Next.js, React, Node.js, Express.js, FastAPI, TailwindCSS, shadcn/ui, Supabase, Firebase
- **Tools & Methodologies:** Git, GitHub, Agile Sprints, Figma, WebSockets, REST APIs, Vercel, Render